

[1] What is the error in the following segment?

```
int x = 10;  
float y = 4.25;  
x = y % x;
```

Errors & Corrections:

- Modulo (%) operator only works on integers.
- y is a float, but % is only valid for int types.
- Solution: Convert y to int before using %:
- $x = ((\text{int})y) \% x;$
- Alternatively, if floating-point remainder is needed, use fmod() from <math.h>:

```
#include <math.h>  
x = fmod(y, x);
```

[2] Errors in the following statements:

(a)

```
if (m==1 & n!=0)
    printf("OK");
```

Error:

Single & is a bitwise operator, not a logical operator.

Solution: Use && instead of &:

```
if (m==1 && n!=0)
    printf("OK");
```

(b)

```
if (x =< 5)
    printf("Jump");
```

Error:

- =< is not a valid relational operator.
- Solution: Use <=:
- if (x <= 5)
- printf("Jump");

[3] Errors in the assignment statements and their corrections:

Original Code	Error	Corrected Code
<code>x = y % x;</code>	<code>%</code> only works with integers	<code>x = ((int)y) % x;</code>
<code>if (m==1 & n!=0)</code>	<code>&</code> should be <code>&&</code>	<code>if (m==1 && n!=0)</code>
<code>if (x =< 5)</code>	<code>=<</code> is incorrect	<code>if (x <= 5)</code>
<code>x = y = z = 0.5, 2.0, -5.75;</code>	Incorrect comma usage	<code>x = 0.5; y = 2.0; z = -5.75;</code>
<code>m = ++a * 5;</code>	<code>a</code> may be undeclared	Declare <code>a</code> before use
<code>y = sqrt(100);</code>	<code>math.h</code> missing	<code>#include <math.h></code>
<code>p *= x/y;</code>	Possible division by zero	<code>if (y != 0) p *= x / y;</code>
<code>s = / 5;</code>	<code>/</code> has no left operand	<code>s = x / 5;</code>
<code>a = b++ - c * 2;</code>	<code>b</code> and <code>c</code> may be undeclared	Declare <code>b</code> and <code>c</code> before use

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